

Geometric Response and State-Resolved Decoherence: Microscopic Born–Markov Derivation for a Commuting Qubit–Bath Coupling

Anderson Costa Rodrigues^{1,*}

¹*Independent Researcher, Maricá, Rio de Janeiro, Brazil*

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We analyze the angular dependence of the purity-loss rate in a restricted class of open quantum systems: a qubit coupled to a thermal bosonic bath through a commuting interaction operator. Within the Born–Markov approximation, we derive the dissipative generator in closed form and establish four results.

First, the restricted induced trace norm of the generator (on the Hermitian traceless subspace) is independent of the coupling angle,

$$\|\mathcal{L}_\theta^{(D)}\|_{1 \rightarrow 1, \mathcal{B}_0} = 4\pi\eta k_B T,$$

and we give a closed-form proof of this fact based on the rotational structure of the coupling family, together with an independent numerical cross-check.

Second, the initial purity-loss rate of a pure reference state depends on the coupling angle through the geometric quantity $\Gamma(\theta) = |\langle\psi|A(\theta)|\psi\rangle|^2$, which for the qubit family $A(\theta) = \cos\theta\sigma_x + \sin\theta\sigma_z$ takes the exact form $\Gamma(\theta) = (e_x \cos\theta + e_z \sin\theta)^2$.

Third, the local curvature of $\Gamma(\theta)$ at its maximum defines a curvature-derived angular scale $\sigma_{\text{geo}} = 1/\sqrt{2}$, universal within this coupling family and independent of every physical parameter of the model; the two forms agree to within $\sim 5\%$ at $|\theta - \mu| = \sigma_{\text{geo}}$ (Appendix E).

Fourth, we quantify explicitly the algebraic (non-exponential) decay of the Markov tail integral for the Ohmic–Lorentzian bath considered here, and we show that a slowly time-dependent coupling angle, $\theta(t)$, admits a well-defined adiabatic–Markovian generator; we then prove that the resulting purity-loss functional is *not* proportional, in general, to the trajectory-integrated Frobenius-norm witness of Ref. [1], identifying the specific obstructions (coherent-precession contamination and a linear-versus-quadratic mismatch in Bloch-vector language) responsible, and we pose as an open problem the narrower question of whether any restricted or reformulated version of the witness admits a well-defined relation to the present results.

These results provide an exact microscopic benchmark for trajectory-dependent effective decoherence models: the angular dependence originates from the geometry of the system–environment coupling and the evolving state, not from an intrinsic angular preference of the bath, while a full reduction of the phenomenological model of Ref. [1] to this microscopic setting remains open.

I. INTRODUCTION

Open quantum systems are commonly described by reduced density operators evolving under master equations. In the Markovian regime, the Lindblad–Gorini–Kossakowski–Sudarshan (GKLS) form provides the most general generator of a completely positive, trace-preserving semigroup [2, 3], and its derivation from a microscopic system–bath Hamiltonian in the weak coupling limit was placed on a rigorous footing by Davies [4] and refined by Dümcke and Spohn [5]. A standard ingredient of that derivation is the secular (rotating-wave) approximation, which discards fast-oscillating cross terms between transitions of different Bohr frequency in order to guarantee complete positivity of the resulting generator [2, 6, 7]. This approximation is well controlled when the relevant Bohr frequencies are well separated, but it necessarily discards information about how the reduced dynamics depends on the relative geometry between the

system’s internal dynamics and its coupling to the environment.

A recent phenomenological model [1] introduced the notion of *biographical irreducibility*: pairs of quantum processes (“quantum twins”) that implement the same unitary and share the same initial and final state, $U_A = U_B$ and $\rho_A = \rho_B$, along distinct intermediate trajectories $K_A \neq K_B$, and that are shown numerically and via bootstrap statistics to decohere to measurably different final states, $\rho'_A \neq \rho'_B$. That work quantifies the effect through a trajectory-integrated Frobenius-norm functional and proposes a photonic experimental protocol, but it does not derive the effect from a microscopic system–bath Hamiltonian; the coupling constant relating decoherence strength to trajectory geometry is introduced phenomenologically. A natural question, left open there, is whether and how such an operational, trajectory-dependent quantity emerges from a standard Born–Markov treatment, and in particular whether it coincides with – or merely correlates with – quantities that a microscopic derivation actually produces.

The present work addresses a narrower but exactly solvable version of this question, at the cost of a severe

* anderson.c.rodrigues@hotmail.com

restriction stated here upfront rather than left implicit: we take the system Hamiltonian to be $H_S(\theta) = \omega_S A(\theta)$, proportional to the coupling operator itself, with $A(\theta) = \cos \theta \sigma_x + \sin \theta \sigma_z$. This is a modeling *choice*, not an independently verifiable physical condition: it trivially forces $[H_S(\theta), A(\theta)] = 0$ (any operator commutes with itself), which in turn removes all coherent precession transverse to the coupling axis and reduces the model to pure dephasing in a fixed eigenbasis, with no unitary dynamics at all. This is what buys exact solvability without a secular approximation (Section III), and it is also what limits the model's reach: none of the closed-form results below is claimed to survive once H_S and $A(\theta)$ are allowed to differ. The structural device of using a single operator that plays a dual role relative to H_S and to the coupling is reminiscent of decoherence-free-subspace and noiseless-subsystem constructions [8] – there too, protection follows from an operator with a special relationship to both the system Hamiltonian and the system–bath coupling – but the mechanisms are not the same: those constructions protect a subspace via a collective symmetry of the coupling to a shared bath, whereas here there is no protected subspace and no symmetry beyond the algebraic coincidence $H_S \propto A$; every state in the full qubit Hilbert space dephases except the two eigenstates of $A(\theta)$ itself. Because different implementations of the same logical operation can traverse different intermediate coupling geometries – the operational premise of Ref. [1], and formally the subject matter of quantum circuit architecture [9] and of the process-tensor framework for multi-time quantum processes [10, 11] – the angular structure derived here is relevant to that broader setting, even though the model itself is restricted to this exactly-solvable special case.

We derive the exact, non-secular, second-order Born–Markov generator for this model (Section III), establish in closed form that its restricted trace norm is independent of the coupling angle (Section VI), and derive the exact geometric form of the state-resolved purity-loss rate and its curvature-derived angular scale (Sections VIII–IX). We then examine what happens when θ is allowed to vary slowly in time (Section X): we state and justify, via the adiabatic Markovian formalism of Albash *et al.* [12], the conditions under which an instantaneous generator is meaningful, and we then confront directly – rather than assume – the question of whether the resulting purity-loss functional reduces to the Frobenius-norm witness of Ref. [1]. We show that it does not, in general, and we identify precisely why. Section XI discusses experimental feasibility, and Section XII situates the restriction to a commuting interaction within the broader literature on non-Markovian dynamics [13–17].

II. MAIN RESULTS

The central results are as follows.

1. The Born–Markov dissipative generator $\mathcal{L}_\theta^{(D)}$ sat-

isfies $\|\mathcal{L}_\theta^{(D)}\|_{1 \rightarrow 1, \mathcal{B}_0} = 4\pi\eta k_B T$ for every θ (Theorem VI.3), proved in closed form and cross-checked numerically (Appendix G).

2. For a pure state $\rho_\psi = |\psi\rangle\langle\psi|$, $\frac{d}{dt}\text{Tr}(\rho_\psi^2) = 2\gamma(\Gamma(\theta) - 1)$, with $\Gamma(\theta) = |\langle\psi|A(\theta)|\psi\rangle|^2$.
3. For the qubit coupling family, $\Gamma(\theta) = L^2 \cos^2(\theta - \mu)$, where $\mu = \text{atan2}(e_z, e_x)$, and the curvature-defined local angular scale is $\sigma_{\text{geo}} = 1/\sqrt{2}$, independent of L , γ , and every bath parameter.
4. The Markov tail integral of the high-temperature Lorentzian bath correlation function used throughout this work decays *algebraically*, not exponentially (Section IV, Appendix D); the full finite-temperature correlation function's tail is not computed here and need not share this exponent. A slowly varying $\theta(t)$ admits a well-defined adiabatic-Markovian generator under conditions stated in Section X, but the resulting integrated purity loss is provably *not* proportional, in general, to the Frobenius-norm trajectory witness W_{bio} of Ref. [1] (Section XB).

III. MICROSCOPIC MODEL

We consider a total Hamiltonian $H_{\text{tot}} = H_S(\theta) + H_B + H_I(\theta)$, with

$$H_S(\theta) = \omega_S A(\theta), \quad (1)$$

$$H_B = \sum_k \omega_k b_k^\dagger b_k, \quad (2)$$

$$H_I(\theta) = A(\theta) \otimes B, \quad (3)$$

where $B = \sum_k g_k (b_k + b_k^\dagger)$, the standard bilinear system–bath coupling of the Caldeira–Leggett class [18]. The coupling operator is

$$A(\theta) = \cos \theta \sigma_x + \sin \theta \sigma_z, \quad (4)$$

satisfying $A^\dagger = A$ and $A^2 = I$, i.e., $A(\theta) = \mathbf{n}_\theta \cdot \boldsymbol{\sigma}$ with $\mathbf{n}_\theta = (\cos \theta, 0, \sin \theta)$ a unit vector.

The defining property of this model is

$$[H_S(\theta), A(\theta)] = 0 \quad \forall \theta, \quad (5)$$

so that in the interaction picture generated by $H_S(\theta)$, $A_I(t) = e^{iH_S(\theta)t} A(\theta) e^{-iH_S(\theta)t} = A(\theta)$ is time-independent. The interaction is therefore pure dephasing in the eigenbasis of $A(\theta)$: writing ρ in that eigenbasis, the dissipator derived below leaves the populations untouched and decays only the coherence, as follows directly from $A^2 = I$ (Appendix C).

Remark III.1 (Scope of the commuting condition). *As already stated in the Introduction, condition (5) is not*

an independent physical assumption but a consequence of the modeling choice $H_S(\theta) = \omega_S A(\theta)$. A physical realization requires an external control (e.g., a static field or a fixed crystal axis) that simultaneously sets both the qubit quantization axis and the dominant noise-coupling channel along the same direction $\mathbf{n}_\theta$; tunable versions of this alignment, rather than a fixed one, are needed to scan θ , which is why Section XI discusses a linear-optical rather than a solid-state realization. This is the price paid for obtaining an exact (non-secular) second-order kernel: the angular dependence isolated below is purely geometric, uncontaminated by rotating-wave corrections that would otherwise appear when $H_S(\theta)$ and $A(\theta)$ do not commute. We do not claim that this alignment persists under generic extensions of the model (e.g., additional qubits or additional coupling channels); it is a hypothesis to be checked case by case, not a generic feature, and Section XII returns to this point.

IV. BATH CORRELATION FUNCTION AND ITS MARKOV TAIL

The environment is initially thermal, $\rho_B = e^{-\beta H_B}/Z_B$. We take an Ohmic spectral density $J(\omega) = \eta \omega e^{-\omega/\omega_c}$. The full bath correlation function $C_{\text{full}}(t) = \langle B(t)B(0) \rangle_{\rho_B} = \int_0^\infty d\omega J(\omega) [\coth(\beta\omega/2) \cos \omega t - i \sin \omega t]$ is complex at any finite temperature; its imaginary part generates, in the standard Redfield/Lindblad reduction, a Lamb-shift correction $H_{\text{LS}} \propto A(\theta)^2$. Because $A(\theta)^2 = I$ for every θ (Section III), H_{LS} is proportional to the identity and therefore commutes with everything: it is dynamically trivial *at any temperature*, not as a consequence of any approximation made below. This is the actual reason the imaginary part of $C_{\text{full}}(t)$ never appears in the generator derived in Section V. What the high-temperature regime $\beta\omega_c \ll 1$ is used for is different and more limited: it gives the specific closed (Lorentzian) form of the *real* part that we use for all explicit calculations below,

$$\begin{aligned} C(t) &\equiv \text{Re } C_{\text{full}}(t) \Big|_{\beta\omega_c \ll 1} = \frac{2\eta}{\beta} \int_0^\infty e^{-\omega/\omega_c} \cos(\omega t) d\omega \\ &= \frac{2\eta\omega_c}{\beta} \frac{1}{1 + \omega_c^2 t^2}. \end{aligned} \quad (6)$$

The time integral required for the Markov generator is

$$\int_0^\infty C(t) dt = \frac{\pi\eta}{\beta}, \quad (7)$$

so the Markov coefficient is finite. The derivation below (Section V) requires this integral to be well approximated by extending its upper limit from a finite system-evolution time t to infinity; the associated truncation error is the tail integral $\int_t^\infty C(s) ds$.

Proposition IV.1 (Algebraic Markov tail). *For the correlation function (6),*

$$\int_t^\infty C(s) ds = \frac{\eta}{\beta} \left[\pi - 2 \arctan(\omega_c t) \right] \xrightarrow{\omega_c t \gg 1} \frac{2\eta}{\beta \omega_c t}, \quad (8)$$

i.e., the tail vanishes only algebraically, as $1/t$, not exponentially.

This is a direct consequence of $C(t)$ being Lorentzian in time: the *exponential* cutoff of the spectral density $J(\omega)$ in frequency space does not translate into an exponential decay of the correlation function in the classical (high-temperature) limit used here. Table I gives the relative tail, $[\int_t^\infty C(s) ds] / [\int_0^\infty C(s) ds]$, at several multiples of the bath correlation time $\tau_B = \omega_c^{-1}$ (numerical values from the reproducibility script, dimensionless units $\eta = \beta = \omega_c = 1$; see Appendix D for the closed form). We stress that $\eta = \beta = \omega_c = 1$ is a convenient normalization for verifying the θ - and parameter-independent *analytic* formulas above – it is not a claim that this specific point satisfies $\beta\omega_c \ll 1$ itself ($\beta\omega_c = 1$ at these values, not $\ll 1$); Table I's percentages are, in any case, ratios of the same integral and are independent of the (η, β, ω_c) point chosen, depending only on $\omega_c t$. Physical parameters for a specific platform must independently satisfy $\beta\omega_c \ll 1$ for Eq. (6) itself to apply; this is checked for the proposed platform in Section XI.

TABLE I. Relative Markov tail $[\int_t^\infty C ds] / [\int_0^\infty C ds]$ as a function of t/τ_B . Convergence is algebraic: an order of magnitude improvement in the truncation error costs roughly an order of magnitude in elapsed bath correlation times, not merely a constant multiple as for an exponentially decaying kernel.

t/τ_B	relative tail
1	50.0%
5	12.6%
10	6.3%
50	1.3%

Remark IV.2. *The Markov extension used in Section V is therefore controlled only for system evolution times $t \gg \tau_B$ with a generous margin: Table I shows that $t = 10\tau_B$ still carries a 6.3% uncorrected tail. A quantitative statement of the Markov approximation's accuracy for this model must reference Table I (or Eq. (8)) directly, specifying the targeted truncation error, rather than invoking the qualitative separation $\tau_B \ll \tau_S$ alone. We stress that algebraic decay does not, by itself, invalidate the Markov approximation: the tail integral still vanishes as $t \rightarrow \infty$ (Eq. (8)), and the approximation is controlled once t is large enough on the quantitative scale set by Table I, not the qualitative one. What algebraic decay changes is how many correlation times are required for a given target accuracy – more than for an exponentially decaying kernel at the same τ_B – which is exactly why*

we report the multiplier explicitly rather than asserting $\tau_B \ll \tau_S$ as if it were self-evidently satisfied.

We also record explicitly, for use throughout, that three distinct approximations are stacked in this derivation: (i) the Born (weak-coupling, second-order-in- H_I) approximation, controlled for $\eta \ll 1$ in the dimensionless units used above; (ii) the Markov approximation, controlled by the separation of time scales $t \gg \tau_B$ (equivalently $\omega_c t \gg 1$), quantified by the algebraic bound of Proposition IV.1 and Table I – *not* by $\gamma t \gg 1$, a numerically similar-looking but logically distinct condition, since γ and ω_c are independent parameters of the model and the two inequalities coincide only if $\gamma \sim \omega_c$; and (iii) the high-temperature approximation $\beta\omega_c \ll 1$, used only to fix the closed (Lorentzian) form (6) of $\text{Re } C_{\text{full}}(t)$, not (as clarified above) to justify dropping the imaginary part. Only (ii) is quantified here in closed form; (i) and (iii) are standard but their combined regime of joint validity, for a specific target platform, must be checked against that platform's actual (η, β, ω_c) , as done in Section XI. The identity $A(\theta)^2 = I$ underlying the Lamb-shift argument above is verified symbolically for all θ in the reproducibility script.

V. BORN-MARKOV DISSIPATIVE GENERATOR

Using the Nakajima–Zwanzig projection formalism [19, 20], the second-order Born equation in the interaction picture is

$$\frac{d\rho(t)}{dt} = - \int_0^t ds \text{Tr}_B [H_I(t), [H_I(t-s), \rho(t-s)\rho_B]]. \quad (9)$$

(The explicit four-step reduction from the exact projector equation to this expression, and the role of the commuting condition (5) in eliminating secular corrections, is given in Appendix A.) Under the Born approximation $\rho_{\text{tot}}(t) \simeq \rho(t) \otimes \rho_B$ and the Markov condition of Section IV, the upper integration limit is extended to infinity. Because $A_I(t) = A$, the dissipative kernel becomes

$$\mathcal{K}_\theta(s)[\rho] = -C(s)[A, [A, \rho]] = 2C(s)(A\rho A - \rho), \quad (10)$$

using $A^2 = I$. Integrating over $s \in [0, \infty)$ yields the Markov generator

$$\mathcal{L}_\theta^{(D)}[\rho] = \gamma(A(\theta)\rho A(\theta) - \rho), \quad (11)$$

with

$$\gamma = 2 \int_0^\infty C(s) ds = \frac{2\pi\eta}{\beta}. \quad (12)$$

Unlike the generic Redfield equation [21], which is a valid second-order Born–Markov generator but is not guaranteed to preserve complete positivity without a

further (typically secular) approximation, the commuting structure of Section III delivers Eq. (11) already in Lindblad–GKLS form [2, 3]: it can be written as $\mathcal{L}_\theta^{(D)}[\rho] = L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}$ with $L = \sqrt{\gamma}A(\theta)$ and $L^\dagger L = \gamma I$, guaranteeing complete positivity.

VI. RESTRICTED INDUCED TRACE NORM OF THE GENERATOR

We define the Hermitian traceless subspace

$$\mathcal{B}_0 = \{X = X^\dagger; \text{Tr } X = 0\}, \quad (13)$$

and the restricted induced trace norm

$$\|\mathcal{L}_\theta^{(D)}\|_{1 \rightarrow 1, \mathcal{B}_0} = \sup_{X \in \mathcal{B}_0 \setminus \{0\}} \frac{\|\mathcal{L}_\theta^{(D)}[X]\|_1}{\|X\|_1}. \quad (14)$$

Lemma VI.1 (Bloch representation [22]). *Every $X \in \mathcal{B}_0$ can be written as $X = \mathbf{r} \cdot \boldsymbol{\sigma}$ for a unique $\mathbf{r} \in \mathbb{R}^3$, and $\|X\|_1 = 2\|\mathbf{r}\|$. Moreover $A(\theta)XA(\theta) = (R_\theta \mathbf{r}) \cdot \boldsymbol{\sigma}$, where $R_\theta = 2\mathbf{n}_\theta \mathbf{n}_\theta^T - I$.*

Proof. $X = \mathbf{r} \cdot \boldsymbol{\sigma}$ has eigenvalues $\pm\|\mathbf{r}\|$ (since $X^2 = \|\mathbf{r}\|^2 I$), so $\|X\|_1 = 2\|\mathbf{r}\|$. For the second statement, use the Pauli identity $(\mathbf{a} \cdot \boldsymbol{\sigma})(\mathbf{b} \cdot \boldsymbol{\sigma}) = (\mathbf{a} \cdot \mathbf{b})I + i(\mathbf{a} \times \mathbf{b}) \cdot \boldsymbol{\sigma}$ twice to expand $A(\theta)XA(\theta) = (\mathbf{n}_\theta \cdot \boldsymbol{\sigma})(\mathbf{r} \cdot \boldsymbol{\sigma})(\mathbf{n}_\theta \cdot \boldsymbol{\sigma})$; a direct expansion (or the standard reflection formula for a unit vector) gives $A(\theta)XA(\theta) = [2(\mathbf{n}_\theta \cdot \mathbf{r})\mathbf{n}_\theta - \mathbf{r}] \cdot \boldsymbol{\sigma} = (R_\theta \mathbf{r}) \cdot \boldsymbol{\sigma}$. $\square$

Lemma VI.2. *$R_\theta - I$ has eigenvalues $\{0, -2, -2\}$ for every θ .*

Proof. $R_\theta \mathbf{n}_\theta = 2\mathbf{n}_\theta(\mathbf{n}_\theta \cdot \mathbf{n}_\theta) - \mathbf{n}_\theta = \mathbf{n}_\theta$, so $\mathbf{n}_\theta$ is an eigenvector of R_θ with eigenvalue $+1$. For any $\mathbf{v} \perp \mathbf{n}_\theta$, $R_\theta \mathbf{v} = 2\mathbf{n}_\theta(\mathbf{n}_\theta \cdot \mathbf{v}) - \mathbf{v} = -\mathbf{v}$, so the two-dimensional subspace orthogonal to $\mathbf{n}_\theta$ is an eigenspace with eigenvalue -1 . Hence R_θ has eigenvalues $\{1, -1, -1\}$, and $R_\theta - I$ has eigenvalues $\{0, -2, -2\}$, independent of θ (the eigenvalues depend only on the fact that $\mathbf{n}_\theta$ is a unit vector, not on its direction). $\square$

Theorem VI.3. *The restricted induced trace norm of the dissipative generator is*

$$\|\mathcal{L}_\theta^{(D)}\|_{1 \rightarrow 1, \mathcal{B}_0} = 2\gamma = \frac{4\pi\eta}{\beta} = 4\pi\eta k_B T \quad (15)$$

for all θ .

Proof. By Lemma VI.1, $\mathcal{L}_\theta^{(D)}[X] = \gamma(R_\theta - I)\mathbf{r} \cdot \boldsymbol{\sigma}$, so $\|\mathcal{L}_\theta^{(D)}[X]\|_1 = 2\gamma\|(R_\theta - I)\mathbf{r}\|$ and $\|X\|_1 = 2\|\mathbf{r}\|$. The supremum of the ratio over $\mathbf{r} \neq 0$ is, by definition, the operator (spectral) norm of the real symmetric matrix $R_\theta - I$, i.e., its largest singular value, which by Lemma VI.2 equals 2 for every θ . Hence $\|\mathcal{L}_\theta^{(D)}\|_{1 \rightarrow 1, \mathcal{B}_0} = 2\gamma$, independent of θ . $\square$

Thus, for the isotropic scalar bath considered, the norm does not select a preferred coupling direction; this is now an exact algebraic identity rather than a numerically observed coincidence.

Remark VI.4 (Operational meaning of $\|\cdot\|_{1 \rightarrow 1, \mathcal{B}_0}$). For any two states ρ_1, ρ_2 , the difference $\rho_1 - \rho_2$ lies in $\mathcal{B}_0$, and the trace distance is $D(\rho_1, \rho_2) = \frac{1}{2} \|\rho_1 - \rho_2\|_1$. Theorem VI.3 therefore bounds the maximal rate at which the dissipative generator can change the trace distance – equivalently, the operational distinguishability – between any two states of the system: $|\dot{D}| \leq \frac{1}{2} \left\| \mathcal{L}_\theta^{(D)} \right\|_{1 \rightarrow 1, \mathcal{B}_0} \|\rho_1 - \rho_2\|_1 / \|\rho_1 - \rho_2\|_1 = 2\pi\eta k_B T$, uniformly in θ . This is the sense in which the restricted norm, rather than the full superoperator norm, is the physically relevant quantity here.

The primary evidence for Theorem VI.3 is the closed-form proof above; as an independent, non-optimization-based check, direct evaluation of $\|\mathcal{L}_\theta^{(D)}[X]\|_1 / \|X\|_1$ on the three basis operators $\{\sigma_x, \sigma_y, \sigma_z\}$ of $\mathcal{B}_0$ reproduces $4\pi\eta k_B T$ to 10^{-10} relative precision at nine representative angles. This is *not*, in general, sufficient to find the maximum of a quadratic-form ratio over $\mathcal{B}_0$; Appendix G explains precisely why it is sufficient here (a structural fact specific to this coupling family, not a general property of quadratic forms) before this evidence is used. A separate, heuristic global optimization (Nelder–Mead, random restarts) is kept only as a further, independent cross-check and is logically subordinate to both of the above (Appendix G).

VII. STATE-RESOLVED PURITY-LOSS RATE

Let $\rho_\psi = |\psi\rangle\langle\psi|$ be a pure state. The purity $P(t) = \text{Tr}(\rho^2)$ evolves as $\frac{dP}{dt} = 2 \text{Tr}(\rho \mathcal{L}_\theta^{(D)}[\rho])$. Using Eq. (11) and $\text{Tr}(\rho A \rho A - \rho^2) = |\langle\psi|A|\psi\rangle|^2 - 1$ (Appendix C),

$$\left. \frac{dP}{dt} \right|_{t=0} = 2\gamma(\Gamma(\theta) - 1), \quad (16)$$

where the *geometric response function* is defined as

$$\Gamma(\theta) = |\langle\psi|A(\theta)|\psi\rangle|^2. \quad (17)$$

$\Gamma(\theta)$ is a state-resolved geometric overlap, not a property of the bath; Theorem VI.3 shows explicitly that the bath contributes only the overall, angle-independent rate γ .

VIII. EXACT GEOMETRIC FORM OF THE RESPONSE

For a qubit state, define $e_x = \langle\sigma_x\rangle$, $e_z = \langle\sigma_z\rangle$. Using Eq. (4), $\langle A(\theta) \rangle = e_x \cos \theta + e_z \sin \theta$, so that

$$\Gamma(\theta) = (e_x \cos \theta + e_z \sin \theta)^2. \quad (18)$$

Let $L^2 = e_x^2 + e_z^2 > 0$ and $\mu = \text{atan2}(e_z, e_x)$, taken in its natural range $(-\pi, \pi]$. No further reduction of μ is needed: $\cos^2(\theta - \mu)$ is already exactly π -periodic in μ , so μ and $\mu + \pi$ give an identical $\Gamma(\theta)$ for every θ without any explicit modular reduction, which would otherwise risk introducing artificial discontinuities were μ ever tracked as a continuously time-dependent quantity. Then

$$\Gamma(\theta) = L^2 \cos^2(\theta - \mu). \quad (19)$$

Proposition VIII.1. The maximum response occurs at $\theta = \mu \pmod{\pi}$; the minimum purity loss occurs when the coupling direction is aligned with the state’s Bloch projection onto the xz -plane.

Remark VIII.2 (Blindness to e_y). Because the coupling family (4) spans only the xz subspace of Pauli operators, $\Gamma(\theta)$ in Eq. (18) depends on the state only through (e_x, e_z) , not through $e_y = \langle\sigma_y\rangle$. A state with $e_x = e_z = 0$ but $e_y \neq 0$ has $\Gamma(\theta) = 0$, and hence the same, maximal, purity-loss rate 2γ for every θ – the angular dependence derived here characterizes only the coherence resolved by the coupling family itself, not the full Bloch vector. This is a limitation of the two-parameter family $A(\theta) = \cos \theta \sigma_x + \sin \theta \sigma_z$ specifically, not of the underlying formalism; a three-parameter coupling family spanning all of $\mathfrak{su}(2)$ would resolve e_y as well, at the cost of the single-angle parametrization used throughout this work.

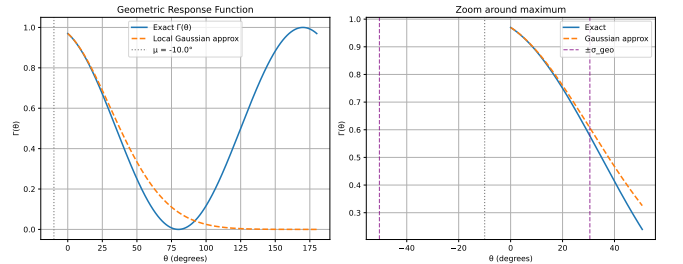


FIG. 1. The exact geometric response function $\Gamma(\theta) = L^2 \cos^2(\theta - \mu)$ (solid) compared with the curvature-matched local approximation $L^2 \exp[-(\theta - \mu)^2 / (2\sigma_{\text{geo}}^2)]$ (dashed), for the representative state $|\psi\rangle = \cos(50^\circ)|0\rangle + \sin(50^\circ)|1\rangle$. Left: full angular range, period π . Right: zoom to $|\theta - \mu| \leq 1.5\sigma_{\text{geo}}$, vertical lines at $\mu \pm \sigma_{\text{geo}}$.

IX. LOCAL CURVATURE AND GEOMETRIC ANGULAR SCALE

The derivatives of Γ at its maximum are

$$\Gamma'(\mu) = 0, \quad \Gamma''(\mu) = -2L^2, \quad \Gamma'''(\mu) = 0. \quad (20)$$

A Gaussian $L^2 \exp[-(\theta - \mu)^2 / (2\sigma^2)]$ has the same second-order Taylor expansion at $\theta = \mu$ if $\sigma^2 = -\Gamma(\mu) / \Gamma''(\mu)$.

We define the *curvature-derived angular scale*

$$\sigma_{\text{geo}} = \frac{1}{\sqrt{2}} \text{ rad} \approx 40.5^\circ. \quad (21)$$

Remark IX.1 (Universality of σ_{geo}). Equation (21) depends on L , e_x , e_z , γ , η , β , and ω_c only through their absence: the ratio $-\Gamma(\mu)/\Gamma''(\mu)$ is fixed by the functional form $\cos^2(\theta - \mu)$ alone, which in turn follows purely from $A(\theta)$ being a linear combination of σ_x and σ_z with unit-norm coefficients. σ_{geo} is therefore not a material property of any particular bath or state; it is a geometric invariant of the coupling family, and its experimental value (Section XI) is a test of the functional form (19) itself, not of any fitted parameter. This scale is not the width of a Gaussian fit to $\Gamma(\theta)$: it is the unique Gaussian whose second-order Taylor expansion matches the cosine squared. The exact function decays more slowly than the Gaussian away from the peak: at $|\theta - \mu| = \sigma_{\text{geo}}$, $\Gamma = \cos^2(1/\sqrt{2})L^2 \approx 0.578 L^2$, while the Gaussian gives $e^{-1/2}L^2 \approx 0.607 L^2$, a fourth-order effect of about 5% (Appendix E) that grows rapidly beyond $1.5 \sigma_{\text{geo}}$, as shown in Fig. 2.

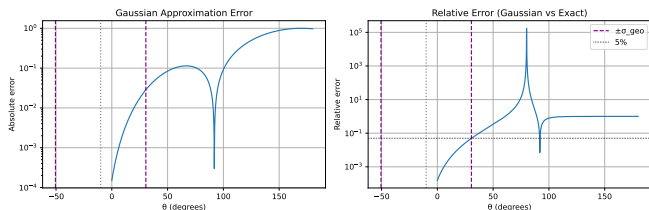


FIG. 2. Absolute (left) and relative (right) deviation between the exact response $\Gamma(\theta)$ and its curvature-matched Gaussian approximation, log scale, same reference state as Fig. 1. The relative error stays below the 5% line (horizontal dotted) inside $|\theta - \mu| \leq \sigma_{\text{geo}}$ (vertical dashed) and grows rapidly beyond it, confirming σ_{geo} delimits a strictly local regime rather than a global fit.

As shown in Fig. 2 and quantified in Appendix E, this 5% figure is itself a parameter-free property of the $\cos^2$ functional form and would apply identically to any two-level system realizing Eq. (19), regardless of the physical origin of the coupling.

X. TIME-DEPENDENT TRAJECTORIES

A. Adiabatic-Markovian validity

The results above hold at fixed θ . Suppose now that θ is allowed to vary slowly in time, $\theta = \theta(t)$, so that $H_S(t) \equiv H_S(\theta(t)) = \omega_S A(\theta(t))$ and $A(t) \equiv A(\theta(t))$ are now genuinely time-dependent operators in the *Schrödinger* picture – a different, externally driven time dependence from the interaction-picture constancy established in Section III, which held only at fixed θ .

Whether an instantaneous generator $\mathcal{L}_{\theta(t)}^{(D)}$, built by simply inserting $\theta(t)$ into Eq. (11), provides a valid description of the driven dynamics is not automatic and requires the adiabatic-Markovian formalism of Albash *et al.* [12], which established, from first principles, the master equation appropriate to a system evolving adiabatically while weakly coupled to a thermal bath. Two ingredients of that formalism are favorable here. First, because $A(\theta)^2 = I$ for every θ , $H_S(\theta) = \omega_S A(\theta)$ has the constant spectral gap $2\omega_S$ for all θ : unlike generic adiabatic protocols, there is no risk of a closing gap invalidating the adiabatic condition at some point along the trajectory. Second, the bath correlation function is unaffected by $\theta(t)$, so only the system side of the adiabatic condition needs to be imposed. Adapting the standard adiabatic theorem together with the time-scale separation of Section IV, a sufficient condition is

$$|\dot{\theta}| \ll \omega_S, \quad |\dot{\theta}| \ll \gamma, \quad |\dot{\theta}| \tau_B \ll 1, \quad (22)$$

i.e., the coupling angle must vary slowly compared to all three time scales of the problem: the (constant) system gap, the dissipative rate, and the bath correlation time. Under condition (22), the instantaneous generator $\mathcal{L}_{\theta(t)}^{(D)}$ of Eq. (11) provides a controlled Markovian description at each instant. Unlike the previous version of this section, we do not merely adapt condition (22) by analogy: Appendix B gives an exact change of frame, tracking the instantaneous eigenbasis of $A(\theta(t))$, under which the naively-substituted generator is shown to be the leading term of a controlled expansion in $|\dot{\theta}|/\omega_S$, with the first condition in Eq. (22) emerging from an explicit rotating-wave argument rather than an assumed analogy; the coefficient of the next-order correction is not computed there, only its parametric suppression. This derivation supports *only* the present subsection – none of Sections III–IX, nor the negative result of Section XB below, rely on it, since those results are all obtained at fixed θ or from Bloch-vector kinematics that hold regardless of the generator's precise form. A separate condition is needed to integrate it: Eq. (16) is an initial rate, derived for a pure state, and $|\psi(t)\rangle$ shrinks toward the interior of the Bloch ball under Eq. (11) itself for any $t > 0$. Extrapolating the $t = 0$ rate linearly,

$$\Delta P \approx 2\gamma \int_0^T (\Gamma(\theta(t); \psi(0)) - 1) dt, \quad (23)$$

is therefore only a *leading-order* approximation, valid for $|\Delta P| \ll \frac{1}{2}$ (the full range available to a qubit), not for arbitrary T . We verified this explicitly (Appendix F): for the worst-case trajectory (maximal decay rate, $\Gamma \equiv 0$), Eq. (23) agrees with the exact solution of the Bloch equation to within 20% for $\gamma T \lesssim 0.2$, but becomes unphysical – predicting purity below the maximally-mixed floor of $\frac{1}{2}$ – beyond $\gamma T \approx 0.3$ (Fig. 3). Using Eq. (23) beyond this regime requires solving the coupled Bloch equation for $\mathbf{r}(t)$ directly rather than integrating the initial rate. We stress that condition (22) and the $|\Delta P| \ll \frac{1}{2}$ condition

just derived are two logically independent requirements – one on how fast $\theta(t)$ may vary, the other on how long the linearized rate may be integrated – and that neither says anything about the relationship between ΔP and any other trajectory functional, which we take up next.

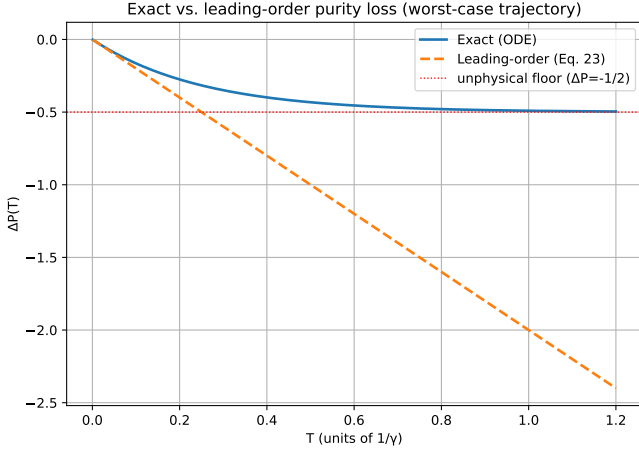


FIG. 3. Exact purity loss (solving the Bloch equation) versus the leading-order approximation of Eq. (23), for the worst-case trajectory of Appendix F ($\Gamma \equiv 0$, maximal decay rate). The leading-order formula is accurate only for $\gamma T \lesssim 0.2$ – 0.3 and becomes unphysical (crosses the $\Delta P = -1/2$ floor, dotted) beyond that.

B. Relation to the biographical-witness functional

Reference [1] quantifies trajectory-dependent decoherence through the Frobenius-norm functional $W_{\text{bio}} = \int_0^T \|\dot{\rho}(t)\|_F dt$, the arc length traced by the density operator in the Hilbert–Schmidt metric, and reports distinguishability between quantum twins via the Frobenius-norm difference of their final states. It is tempting to identify Eq. (23) as a microscopic derivation of W_{bio} ; we show here that this identification does *not* hold in general, and we isolate the obstruction explicitly rather than leave it implicit. The argument below depends on none of the machinery of Section XA: it uses only the general, model-independent Bloch-vector identities (24), which hold for *any* qubit dynamics, and is stated pointwise in time rather than as an integrated trajectory quantity, so it is unaffected by whatever remains open in the adiabatic reduction of Appendix B.

Write $\rho = (I + \mathbf{r} \cdot \boldsymbol{\sigma})/2$ for a qubit. A short calculation (Appendix F) gives, for *any* qubit dynamics,

$$\|\dot{\rho}\|_F = \frac{|\dot{\mathbf{r}}|}{\sqrt{2}}, \quad \frac{dP}{dt} = \mathbf{r} \cdot \dot{\mathbf{r}}. \quad (24)$$

The first quantity is the instantaneous *speed* of the Bloch vector; the second is its instantaneous *radial* component of motion (purity depends only on $|\mathbf{r}|$). These coincide only when $\dot{\mathbf{r}}$ is parallel to $\mathbf{r}$ at every instant – i.e., when

the trajectory is a purely radial contraction with no tangential (purity-preserving) component.

Proposition X.1 (Non-proportionality). ΔP [Eq. (23)] is not proportional to W_{bio} in general. This is a proven negative result, not an open question: the two obstructions below establish it directly, one of which suffices.

Obstruction 1 (coherent contamination). W_{bio} is defined in Ref. [1] for arbitrary open-system trajectories, not only commuting ones; testing it against a trajectory with a coherent component is therefore squarely within its intended scope, even though such a component falls outside the commuting restriction that defines the microscopic model of Sections III–IX. The Bloch-vector identities (24) used below hold for any qubit dynamics, commuting or not. If the trajectory includes any coherent (unitary) component – for instance residual precession from $H_S(\theta(t))$ itself, or any control imperfection – that component contributes to $|\dot{\mathbf{r}}|$, and hence to W_{bio} , while contributing *exactly zero* to dP/dt , since unitary evolution conserves purity. Concretely, for dephasing along a fixed axis ($\theta = 0$) with an added precession $H_S = (\omega_0/2)\sigma_z$ – which explicitly breaks $[H_S, A] = 0$, illustrating the general case rather than a property internal to the commuting model – acting on an equatorial state, one finds $dP/dt = -2\gamma$ for *every* value of ω_0 , while $\|\dot{\rho}\|_F$ grows without bound as ω_0 increases (a factor of 25 between $\omega_0 = 0$ and $\omega_0 = 50\gamma$ in the example worked in Appendix F). W_{bio} is therefore sensitive to coherent dynamics that ΔP cannot see at all.

Obstruction 2 (linear vs. quadratic dependence). Even in the strictly dissipative sub-case with no coherent term at all, and with the Bloch vector confined to the plane transverse to the coupling axis ($\mathbf{r} = \mathbf{r}_\perp$, purely radial decay, $\dot{\mathbf{r}} = -2\gamma\mathbf{r}_\perp$), Eq. (24) gives

$$\|\dot{\rho}\|_F = \sqrt{2}\gamma|\mathbf{r}_\perp|, \quad \frac{dP}{dt} = -2\gamma|\mathbf{r}_\perp|^2, \quad (25)$$

linear and quadratic in $|\mathbf{r}_\perp|$, respectively (Appendix F). Their ratio is $-\sqrt{2}|\mathbf{r}_\perp|$, which is not constant along a trajectory on which $|\mathbf{r}_\perp|$ itself decays. So even in the most favorable case – no coherent term, and motion restricted to the purely radial direction – W_{bio} and ΔP are related by a state-dependent, nonlinear map, not a proportionality constant.

We conclude that Eq. (23) is a rigorous microscopic result for the purity-loss functional in its own right, and it is *consistent with* the qualitative claim of Ref. [1] that trajectory geometry affects decoherence outcomes, but it does not constitute a derivation of W_{bio} or of the phenomenological relation between decoherence strength and trajectory length proposed there.

Open Problem X.2. Proposition X.1 rules out the naive identification $\Delta P \propto W_{\text{bio}}$; it does not rule out every possible relation. Is there a restriction of W_{bio} (e.g., to its radial component) or a reformulation of the phenomenological model of Ref. [1] directly in terms of $\Gamma(\theta(t))$ under

which a well-defined, non-trivial relation to the present microscopic results can be established?

XI. EXPERIMENTAL FEASIBILITY

The static and dynamic results of this paper require different experimental platforms, and conflating them would overstate what a single setup can test. We treat them separately.

A. Static-angle predictions (linear optics)

The predictions below concern the coupling-angle dependence of Sections VIII–IX at fixed θ , which does not require the adiabatic machinery of Section X.

1. **Peak direction.** For any prepared pure state, the coupling angle of minimum purity loss must be $\mu = \text{atan2}(e_z, e_x)$; varying the initial state tests this relation directly.
2. **Angular profile.** The purity-loss rate must follow $\cos^2(\theta - \mu)$, with curvature ratio $\Gamma''(\mu)/\Gamma(\mu) = -2$ (Remark IX.1), independent of state, temperature, and coupling strength.
3. **Minimum.** The minimum of Γ occurs at $\mu + \pi/2$.

A natural platform is a photonic polarization qubit, with θ set by a wave-plate angle and a controllable dephasing channel implemented interferometrically – e.g., a displaced Sagnac interferometer of the kind already used to engineer amplitude-damping and dephasing channels on polarization qubits under waveplate control in weak-measurement and measurement-reversal experiments. A wave plate applies a fixed, essentially instantaneous unitary at a localized point in the optical path; this is entirely adequate for setting a *constant* θ per run and scanning it across runs, which is all predictions 1–3 require.

Two quantitative constraints from the microscopic theory bound the feasible parameter regime. First, Section IV requires the system evolution (interrogation) time to satisfy $t \gtrsim 10\text{--}50 \tau_B$ for the Markov truncation error to fall in the few-percent range (Table I); for an engineered interferometric bath, τ_B is set by the interferometer’s own delay and is therefore, unlike a solid-state phonon bath, a design parameter that can be made short compared to the achievable interrogation time. Second, resolving the 5% deviation between $\Gamma(\theta)$ and its Gaussian approximation (Remark IX.1) at $\theta = \mu \pm \sigma_{\text{geo}}$ requires angular control and readout precision below this level; motorized wave-plate stages with sub-degree repeatability, already standard in polarization-qubit process tomography, are more than adequate, since $\sigma_{\text{geo}} \approx 40.5^\circ$ is a coarse angular scale on that scale. We do not attempt a full experimental design here; the point is that the constraints imposed by Sections IV and IX are quantitative

and platform-independent, and should be checked explicitly against any specific realization rather than assumed.

B. Dynamic/trajectory predictions: a different platform is needed

The adiabatic-Markovian regime of Section X requires $\theta(t)$ to vary *continuously in time while the dissipative generator acts concurrently*, on the time scales set by condition (22). A single wave plate in a linear-optical, single-pass architecture does not realize this: it is a localized, essentially instantaneous gate per photon, not a continuously driven Hamiltonian $H_S(\theta(t))$ acting in parallel with an open-system bath over an extended interrogation window. Reference [1] proposes a photonic protocol for its own trajectory-dependent effect, which is not, on the same grounds, obviously compatible with a static per-photon wave-plate setting either; we do not assume here that it resolves this mismatch, and we do not claim that the static platform of Section XIA tests the dynamic content of Section X. Testing the adiabatic regime and Open Problem X.2 directly calls for a platform with genuine continuous-time Hamiltonian control concurrent with dissipation – for instance a driven superconducting qubit or an NV center, where $\theta(t)$ could be realized as a continuously modulated control field – which we leave to future work rather than retrofit onto the static proposal above.

XII. DISCUSSION

We have derived, from a microscopic model with commuting qubit–bath coupling, an exact relation between the initial purity-loss rate and the geometric overlap $\Gamma(\theta)$, with the angle-independence of the dissipative generator’s norm (Theorem VI.3) now established in closed form rather than by numerical search alone. The cosine-squared form and the curvature scale $1/\sqrt{2}$ are universal within this coupling family (Remark IX.1).

This work does not claim to derive trajectory dependence in full generality, nor – see Section X B – does it claim to derive the phenomenological witness of Ref. [1] from first principles; it isolates a minimal, exactly solvable setting in which the angular mechanism available to any such model can be made precise, and it identifies specifically where a bridge to the phenomenological model would need to be built. When the coupling operator is externally varied, the effective decoherence naturally becomes history-dependent; this is consistent with the process-tensor framework [10, 11] and with the broader literature on non-Markovian quantum dynamics [13–15].

We note explicitly that $\Gamma(\theta)$ is *not* an instance of the geometric (Berry) phase acquired by an adiabatically evolving open system, a distinct and more established notion of “geometric” decoherence in which a slowly cy-

cled Hamiltonian parameter picks up a holonomy corrected by the bath [23]; nor is it a Loschmidt-echo-type fidelity susceptibility to a Hamiltonian perturbation [24]. $\Gamma(\theta)$ is simply the squared overlap of a fixed reference state with the coupling operator itself, evaluated at fixed θ ; the word “geometric” here refers only to this overlap structure on the Bloch sphere, not to a holonomy or a topological invariant. The bath correlation function used in Section IV, and the resulting pure-dephasing rate γ , are also the objects of the filter-function and noise-spectroscopy formalism used to characterize and mitigate qubit dephasing experimentally [25]; realized dephasing times well in excess of $100\mu\text{s}$ alongside $T_1 \sim 10\mu\text{s}$ have been demonstrated for a driven superconducting qubit using exactly this class of technique [26], which is the concrete precedent behind the solid-state platform proposed for the dynamic regime in Section XI.

The restriction to a commuting interaction (Remark III.1) limits the immediate applicability of the exact results of Sections V–IX: they hold only so long as the system’s free-evolution axis remains locked to the coupling geometry. We have not verified that this alignment survives under generic extensions of the model – for instance, additional qubits, additional coupling channels, or coupling operators that are not simple linear combinations of two Pauli matrices – and we explicitly flag this as untested rather than assumed to generalize. The geometric structure identified here (a $\cos^2$ response with a universal, parameter-free curvature scale) is a feature of the two-dimensional coupling family (4) specifically; a non-commuting or higher-dimensional extension would need to be re-derived, likely under a secular approximation, and there is no guarantee that the same closed forms survive that step.

Future work should extend the analysis to non-commuting couplings and to mixed initial states, address Open Problem X.2, and examine the connection with process-tensor-based measures of non-Markovianity.

XIII. CONCLUSION

For a qubit coupled to a thermal Ohmic bath through a commuting interaction $A(\theta)$, the state-resolved purity-loss rate exhibits an exact $\cos^2(\theta - \mu)$ dependence, and the norm of the Born–Markov dissipative generator is provably constant in θ (Theorem VI.3), isolating the geometric origin of the angular modulation. The curvature-defined angular scale $\sigma_{\text{geo}} = 1/\sqrt{2}$ is a universal, parameter-free invariant of the coupling family. The Markov approximation underlying these results converges only algebraically (Proposition IV.1), a quantitative constraint on any proposed realization. A slowly varying coupling angle admits a well-defined adiabatic-Markovian generator, but we prove that its integrated purity loss does not, in general, coincide with the trajectory-integrated Frobenius-norm witness of the phenomenological model it was meant to underwrite (Proposition X.1);

whether any restricted or reformulated version of that witness does is stated as an explicit, narrower open problem rather than papered over. The results above stand on their own as an exact microscopic benchmark for state-resolved, angle-dependent decoherence, independent of that open problem’s resolution.

ACKNOWLEDGMENTS

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Appendix A: Explicit Nakajima–Zwanzig to Markov reduction

This appendix makes explicit the steps compressed into Eqs. (9)–(12).

Step 1 (projectors). Define $\mathcal{P}X = \text{Tr}_B(X) \otimes \rho_B$ and $\mathcal{Q} = \mathcal{I} - \mathcal{P}$. The exact reduced dynamics obeys the Nakajima–Zwanzig equation

$$\dot{\rho}(t) = -i \text{Tr}_B[H_I(t), \rho_{\text{tot}}(t)] - \int_0^t ds \text{Tr}_B[H_I(t), e^{(t-s)\mathcal{Q}\mathcal{L}} \mathcal{Q}\mathcal{L}[H_I(s), \rho(s) \otimes \rho_B]], \quad (\text{A1})$$

where $\mathcal{L}X = -i[H_{\text{tot}}, X]$. The first term vanishes because $\text{Tr}_B[A(\theta), \rho_B B] = A(\theta)\rho \text{Tr}_B(\rho_B B) = 0$ for a thermal bath ($\langle B \rangle_{\rho_B} = 0$).

Step 2 (Born approximation). To second order in H_I , replace $e^{(t-s)\mathcal{Q}\mathcal{L}} \rightarrow e^{(t-s)\mathcal{L}_B}$ (free bath propagation) and $\rho(s) \rightarrow \rho(t)$ (slowly varying system, valid for $\tau_B \ll \tau_S$; see Section IV for the quantitative version of this condition). This gives Eq. (9).

Step 3 (commuting interaction picture). Because $[H_S(\theta), A(\theta)] = 0$ (Eq. (5)), the interaction-picture coupling operator is time-independent, $A_I(t) = A(\theta)$. Only the bath operator carries time dependence. Expanding the double commutator and tracing over the bath using $\text{Tr}_B[\rho_B B(0)B(s)] = C(s)$ yields Eq. (10).

Step 4 (Markov limit). Since $\int_0^\infty |C(s)| ds = \pi\eta/\beta < \infty$ (Eq. (7)), the upper limit is extended to infinity with the error quantified in Proposition IV.1 and Table I. Using $A^2 = I$, $[A, [A, \rho]] = 2(\rho - A\rho A)$, giving the time-independent Lindblad generator of Eqs. (11)–(12). No secular approximation is required at any step, because Eq. (5) guarantees a single time-independent coupling operator throughout; this is the structural simplification purchased by the commuting-coupling restriction (Remark III.1).

Appendix B: Rotating-frame derivation of the adiabatic condition

This appendix derives condition (22) rather than assuming it by analogy, for the specific case of Section X. It supports only that section; it is not used elsewhere.

Exact change of frame. Let $V(t) = e^{i\theta(t)\sigma_y/2}$, which satisfies $V^\dagger A(\theta(t))V = \sigma_x$ for every t (a direct consequence of $A(\theta)$ being confined to the xz -plane, verified symbolically). Define $\tilde{\rho}(t) = V(t)^\dagger \rho(t) V(t)$. Since $V^\dagger \dot{V} = i\dot{\theta}\sigma_y/2$ and $\dot{V}^\dagger V = -i\dot{\theta}\sigma_y/2$, the product rule gives, *exactly* (no approximation yet),

$$\dot{\tilde{\rho}} = -i[\omega_S \sigma_x + \frac{\dot{\theta}}{2} \sigma_y, \tilde{\rho}] + \gamma(\sigma_x \tilde{\rho} \sigma_x - \tilde{\rho}), \quad (\text{B1})$$

starting from the naively-substituted lab-frame equation $\dot{\rho} = -i[H_S(\theta(t)), \rho] + \mathcal{L}_{\theta(t)}^{(D)}[\rho]$ (both sides were checked to agree identically with SymPy, for a general ρ , before writing this). Equation (B1) shows that in this frame the dissipator is the fixed, θ -independent σ_x -dephasing generator of Sections V–VI for *every* trajectory $\theta(t)$: all of the time-dependence has been moved into a single extra coherent term, $(\dot{\theta}/2)\sigma_y$, which is exactly the obstruction, since $[\sigma_x, \sigma_y] = 2i\sigma_z \neq 0$: it is this term, and only this term, that reintroduces the non-commuting structure the rest of the paper is built to avoid.

Why $|\dot{\theta}| \ll \omega_S$ suppresses it. In the interaction picture with respect to the dominant part $\omega_S \sigma_x$,

$$\begin{aligned} \sigma_y(\tau) &= e^{i\omega_S \sigma_x \tau} \sigma_y e^{-i\omega_S \sigma_x \tau} \\ &= \cos(2\omega_S \tau) \sigma_y - \sin(2\omega_S \tau) \sigma_z, \end{aligned} \quad (\text{B2})$$

verified symbolically, i.e., the extra term oscillates at the *constant* gap frequency $2\omega_S$ and averages to zero over any window long compared to $1/\omega_S$ but short compared to the timescale $1/|\dot{\theta}|$ over which its prefactor changes. This is exactly the standard rotating-wave argument, now applied to the extra term rather than to the dephasing structure itself, and it is precisely what requires $|\dot{\theta}| \ll \omega_S$: the prefactor must vary slowly compared to the oscillation it multiplies. Under this condition, the extra term's leading effect on the coarse-grained (Markovian) dynamics of $\tilde{\rho}$ averages away, recovering Eq. (11) $_{\theta \rightarrow \theta(t)}$ as the leading-order result. We have not computed the coefficient of the next-order correction (organized in powers of $|\dot{\theta}|/\omega_S$), only demonstrated the mechanism that suppresses it and the parameter it is suppressed by; the two remaining conditions in Eq. (22) continue to be adapted by analogy with the Markov and bath-memory constraints of Section IV, not independently rederived here.

Appendix C: Purity decay formula

Derivation of Eq. (16): using Eq. (11) and $A^2 = I$,

$$\begin{aligned} \text{Tr}(\rho \mathcal{L}_\theta^{(D)}[\rho]) &= \gamma \text{Tr}(\rho A \rho A - \rho^2) \\ &= \gamma(|\langle \psi | A | \psi \rangle|^2 - 1), \end{aligned} \quad (\text{C1})$$

where the last step uses $\rho = |\psi\rangle\langle\psi|$, so $\text{Tr}(\rho A \rho A) = \langle \psi | A | \psi \rangle \langle \psi | A | \psi \rangle = |\langle \psi | A | \psi \rangle|^2$. In the eigenbasis of $A(\theta)$ (eigenvalues ± 1), writing ρ with entries ρ_{ij} , one has $A\rho A - \rho = \text{diag}(0,0)$ off the diagonal replaced by $-2\rho_{01}, -2\rho_{10}$: populations are unaffected and only coherences decay, confirming the pure-dephasing character of the model stated in Section III.

Appendix D: Algebraic Markov tail: closed form

The symbolic tail integral used for Proposition IV.1 and Table I, verified with SymPy, is

$$\begin{aligned} \int_t^\infty C(s) ds &= \frac{2\eta\omega_c}{\beta} \int_t^\infty \frac{ds}{1 + \omega_c^2 s^2} \\ &= \frac{\eta}{\beta} \left[\pi - 2 \arctan(\omega_c t) \right]. \end{aligned} \quad (\text{D1})$$

Using $\arctan(x) = \pi/2 - 1/x + O(x^{-3})$ for large x gives the algebraic large- t behavior quoted in Eq. (8). The numerical values in Table I follow by dividing by $\int_0^\infty C ds = \pi\eta/\beta$ and setting $\eta = \beta = \omega_c = 1$.

Appendix E: Local Gaussian approximation error

Expanding $\cos^2 x = 1 - x^2 + \frac{x^4}{3} + O(x^6)$ and $e^{-x^2} = 1 - x^2 + \frac{x^4}{2} + O(x^6)$, the difference is $-\frac{x^4}{6} + O(x^6)$. At $x = x_{\text{geo}} = 1/\sqrt{2}$, the absolute difference (from the full, non-truncated expressions, as computed by the reproducibility script) is ≈ 0.0286 . This figure is reported here relative to two different quantities, which must not be conflated: relative to the *local* exact value at that point, $\Gamma(\mu + \sigma_{\text{geo}}) \approx 0.578 L^2$, it is $\approx 4.95\%$; relative to the *peak* value $\Gamma(\mu) = L^2$, it is $\approx 2.86\%$. Both are printed explicitly, and labeled, by the reproducibility script; the main text (Remark IX.1) and Fig. 2 use the local figure, since that is the relevant one for judging the accuracy of the Gaussian approximation *at* the point being evaluated, not relative to a different point (the peak) on the curve.

Appendix F: Bloch-vector calculations for Section X B

For $\rho = (I + \mathbf{r} \cdot \boldsymbol{\sigma})/2$, $\dot{\rho} = (\dot{\mathbf{r}} \cdot \boldsymbol{\sigma})/2$. Since $\text{Tr}(\sigma_i \sigma_j) = 2\delta_{ij}$, $\|\dot{\rho}\|_F^2 = \text{Tr}(\dot{\rho}^2) = \frac{1}{4} \text{Tr}[(\dot{\mathbf{r}} \cdot \boldsymbol{\sigma})^2] = \frac{1}{4} (2|\dot{\mathbf{r}}|^2) = |\dot{\mathbf{r}}|^2/2$, giving the first identity in Eq. (24). Purity is $P = \text{Tr}(\rho^2) = (1 + |\mathbf{r}|^2)/2$, so $dP/dt = \mathbf{r} \cdot \dot{\mathbf{r}}$, the second identity.

Obstruction 1, worked example. Take $\theta = 0$ ($A = \sigma_x$, dephasing preserves the x -component) with $H_S = (\omega_0/2)\sigma_z$ added. The Bloch equations are $\dot{r}_x = -\omega_0 r_y$, $\dot{r}_y = \omega_0 r_x - 2\gamma r_y$, $\dot{r}_z = -2\gamma r_z$. At the equatorial state $\mathbf{r} = (0, 1, 0)$, $\dot{\mathbf{r}} = (-\omega_0, -2\gamma, 0)$, so $dP/dt = \mathbf{r} \cdot \dot{\mathbf{r}} = -2\gamma$ for every ω_0 , while $\|\dot{\rho}\|_F = \sqrt{(\omega_0^2 + 4\gamma^2)/2}$ grows without bound. Numerically, $\|\dot{\rho}\|_F/\gamma \approx 1.41, 1.58, 7.21, 35.4$

at $\omega_0/\gamma = 0, 1, 10, 50$, respectively, while $dP/dt = -2\gamma$ throughout.

Obstruction 2, worked example. For purely radial decay ($\mathbf{r} = \mathbf{r}_\perp$, $\dot{\mathbf{r}} = -2\gamma\mathbf{r}_\perp$), Eq. (24) gives $\|\dot{\rho}\|_F = \sqrt{2}\gamma|\mathbf{r}_\perp|$ and $dP/dt = -2\gamma|\mathbf{r}_\perp|^2$ directly by substitution.

Eq. (23), worst-case worked example. Take $\theta = 0$ fixed ($\mathbf{n} = \hat{\mathbf{x}}$) and the initial state $\mathbf{r}(0) = (0, 1, 0)$, so $\Gamma(0) = 0$ throughout (the $\hat{\mathbf{x}}$ -component of $\mathbf{r}$ never changes at fixed θ , and starts at zero): this is the maximal-decay-rate case, the fastest that ΔP can accumulate for this model. The exact Bloch equation $\dot{\mathbf{r}} = \gamma(R_0 - I)\mathbf{r}$ integrates in closed form, $\mathbf{r}_\perp(t) = \mathbf{r}_\perp(0)e^{-2\gamma t}$, giving $\Delta P_{\text{exact}}(T) = (e^{-4\gamma T} - 1)/2$, while Eq. (23) gives $\Delta P_{\text{naive}}(T) = -2\gamma T$ (using $\Gamma \equiv 0$). Table II compares the two (values from the reproducibility script, direct ODE integration, $\gamma = 1$).

TABLE II. Exact vs. leading-order [Eq. (23)] purity loss for the worst-case trajectory of this appendix. Eq. (23) is unphysical ($\Delta P < -1/2$) beyond $\gamma T \approx 0.3$.

γT	ΔP_{exact}	ΔP_{naive} (Eq. (23))
0.05	-0.091	-0.100
0.10	-0.165	-0.200
0.20	-0.275	-0.400
0.30	-0.349	-0.600 (unphysical)
0.50	-0.432	-1.000 (unphysical)

Appendix G: Numerical cross-checks of Theorem VI.3

The reproducibility script performs two independent numerical checks, neither of which is the primary evi-

dence for Theorem VI.3 (the closed-form proof in Section VI is). First, exact evaluation of $\|\mathcal{L}_\theta^{(D)}[X]\|_1/\|X\|_1$ on the three basis operators $\{\sigma_x, \sigma_y, \sigma_z\}$ of $\mathcal{B}_0$, at nine values of $\theta \in [0, \pi]$. This is not, in general, sufficient to find the maximum of a quadratic-form ratio over $\mathcal{B}_0$ – extrema of such ratios are generally attained only on the eigenbasis of the associated operator, not on an arbitrary spanning set. It is sufficient *here* for a structural reason specific to this coupling family: since $\mathbf{n}_\theta = (\cos \theta, 0, \sin \theta)$ has zero y -component for every θ (Eq. (4) involves only σ_x, σ_z), $\hat{\mathbf{y}}$ is orthogonal to $\mathbf{n}_\theta$ for every θ , and by Lemma VI.2 the 2-dimensional eigenspace of $R_\theta - I$ with the maximal eigenvalue -2 is exactly the plane orthogonal to $\mathbf{n}_\theta$. Hence σ_y alone is guaranteed to lie in that maximal eigenspace for every θ , and evaluating on it already attains the exact supremum; σ_x, σ_z are included only for redundancy. This argument is specific to a coupling family confined to the xz -plane and would need to be redone – most directly, by evaluating on the actual eigenbasis of $R_\theta - I$ rather than on a fixed θ -independent triple – for any extension in which $\mathbf{n}_\theta$ acquires a y -component; all 27 evaluations reproduce 2γ to 10^{-10} relative precision. Second, as a further, method-independent check, a global numerical search for $\sup_X \|\mathcal{L}_\theta^{(D)}[X]\|_1/\|X\|_1$ over $\mathbf{r} \in S^2$, at the same nine angles, using Nelder–Mead with 20 random restarts per angle (tolerance 10^{-12}): all nine maxima agree with 2γ to within 10^{-8} relative precision. Both are consistent with, and logically subordinate to, the closed-form result of Theorem VI.3.

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